

DSA - CSE102 - Mid-Sem Exam

Total Marks : 75 Marks

Date: 23/02/25

Instructions:

1. All the TAs will strictly follow the rubric provided to them. **No requests will be entertained related to the scoring strategy.**
2. The marks are graded based on the most efficient solution to each problem.

Question 1. The recursive model is $T(n) = 8T(n/2) + \Theta(1)$. Derive the time complexity for this model using the recursive tree method and validate using the substitution method. (25 Marks)

Question 2. Given a singly linked list, write a pseudocode to split it into two roughly equal halves. Provide an efficient solution. (25 Marks)

Question 3. You are given a sorted array that has been rotated at some unknown pivot. For example, the sorted array $[1, 2, 3, 4, 5, 6, 7]$ might be rotated to $[4, 5, 6, 7, 1, 2, 3]$. Explain how binary search may be modified to efficiently search for a given target element in such a rotated sorted array. Give pseudocode for this modified binary search and briefly explain your approach. (25 Marks)